

Thm. Let $R_0 > 0$ be given.

Suppose that f is analytic on the upper-half plane outside of $R_0 e^{i\theta}$ ($0 \leq \theta \leq \pi$), and for $C_R: t \mapsto R \cdot e^{it\theta}$ ($0 \leq \theta \leq \pi$) where $R > R_0$, f satisfies

there is M_R such that $\sup_{z \in C_R} |f(z)| \leq M_R$ for all $R > R_0$
and $\lim_{R \rightarrow \infty} M_R = 0$

Then, for all $a > 0$,

$$\lim_{R \rightarrow \infty} \int_{C_R} f(z) \cdot e^{iaz} dz = 0.$$

Proof. First, we will show the Jordan inequality:

Since $\sin x \geq \frac{2}{\pi} \cdot x$ for all $x \in [0, \frac{\pi}{2}]$, therefore
 $e^{-R \sin \theta} \leq e^{-R \cdot \frac{2}{\pi} x}$ for all $x \in [0, \frac{\pi}{2}]$.

The e^{-cx} is integrable explicitly, so

$$\int_0^{\frac{\pi}{2}} e^{-R \sin \theta} d\theta \leq \int_0^{\frac{\pi}{2}} e^{-R \cdot \frac{2}{\pi} x} dx = \frac{\pi}{2R} (1 - e^{-R}) \leq \frac{\pi}{2R}.$$

So, by the symmetry of $e^{-R \sin \theta}$, we obtain the Jordan inequality

$$\int_0^{\pi} e^{-R \sin \theta} d\theta \leq \frac{\pi}{R}. \quad \text{Jordan's inequality.}$$

Now,

$$\int_{C_R} f(z) e^{iaz} dz = \int_0^{\pi} \underbrace{f(Re^{i\theta})}_{|f| \leq M_R} \cdot \underbrace{e^{ia \cdot Re^{i\theta}}}_{|e^{ia \cdot Re^{i\theta}}| \leq e^{-aR \sin \theta}} \cdot \underbrace{R \cdot ie^{i\theta}}_{|R \cdot ie^{i\theta}| = R} d\theta$$

, so that

$$\left| \int_{C_R} f(z) e^{iaz} dz \right| \leq M_R \cdot R \cdot \int_0^{\pi} e^{-aR \sin \theta} d\theta < \frac{M_R \pi}{a} \rightarrow 0 \text{ as } R \rightarrow \infty. \quad \square$$